Basis And Dimension

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Dear students, imagine you're building a file compression algorithm. You want to represent a large dataset (like images or motion data) using the **smallest number of unique pieces**

Kind of like using the least number of building blocks to recreate any shape in LEGO.

That smallest collection of essential pieces is called a **basis**, and the number of them is the **dimension**.

Basis

- A **basis** is a **minimum set of independent vectors** that can **span the entire space**.
- Like the **primary colors** (RGB) can be used to form **any color** on your screen — and none of them is redundant.

A **basis** is both:

- **Spanning**: can create everything in the space
- **Linearly Independent**: no vector is a combination of the others.

Mathematical Definition

Basis: A set of vectors $\{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}, \dots, \overrightarrow{v_k}\}$ in a vector space V is a **basis** if:

- The vectors are **linearly independent.**
- They **span** the space V.

Dimension:

The **dimension** of a vector space is the number of vectors in **any basis** of the space.

Example: Standard Basis

Vectors $\mathrm{v}_1 = (1,0,0)$, $v_2 = (0,1,1), v_3 = (0,0,1)$ form a basis for $\mathbb{R}^3$ as

- They are **linearly independent**
- They **span** $\mathbb{R}^3$ as any vector (x, y, z) can be written as $x\,\overrightarrow{v_1} + y\,\overrightarrow{v_2} + z\,\overrightarrow{v_3}$
- Dimension = 3

Example

Example: Show that the vectors $v_1 = (1,2,1), v_2 = (2,9,0), v_3 = (3,3,4)$ form basis for R^3.

For it we must show that the vectors are linearly independent and span R^3.

Linearly independent:

$$k_1v_1 + k_2v_2 + k_3v_3 = 0$$

$$k_1(1,2,1) + k_2(2,9,0) + k_3(3,3,4) = (0,0,0)$$

$$(k_1 + 2k_2 + 3k_{3,} 2k_1 + 9k_2 + 3k_3, k_1 + 4k_3) = (0,0,0)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 9 & 3 & 0 \\ 1 & 0 & 4 & 0 \end{array}\right)$$

By using Gaussian Jordan method, we come up with

$$k_1 = 0, k_2 = 0, k_3 = 0$$

Hence vectors are linearly independent

Now, we prove that $Span\{v_1, v_2, v_3\} = R^3$

$$(a, b, c) = k_1 v_1 + k_2 v_2 + k_3 v_3$$

$$(a, b, c) = k_1(1,2,1) + k_2(2,9,0) + k_3(3,3,4)$$

$$(a, b, c) = \left(k_1 + 2k_2 + 3k_{3,} 2k_1 + 9k_2 + 3k_3, k_1 + 4k_3\right)$$

$$k_1 + 2k_2 + 3k_3 = a$$

$$2k_1 + 9k_2 + 3k_3 = b$$

$$k_1 + 4k_3 = c$$

Example: Solution

$$\begin{bmatrix} 1 & 2 & 3 & a \\ 2 & 9 & 3 & b \\ 1 & 0 & 4 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 5 & -3 & b-2a \\ 0 & -2 & 1 & c-a \end{bmatrix} \quad \begin{matrix} R_2 - 2R_1 \\ R_3 - R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 1 & -3/5 & (b-2a)/5 \\ 0 & -2 & 1 & c-a \end{bmatrix} \quad \frac{1}{5}R_2$$

$$\begin{bmatrix} 1 & 0 & 21/5 & (9a-2b)/5 \\ 0 & 1 & -3/5 & (b-2a)/5 \\ 0 & 0 & -1/5 & (-9a+2b+5c)/5 \end{bmatrix} \quad \begin{matrix} R_1 - 2R_2 \\ R_3 + R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 21/5 & (9a-2b)/5 \\ 0 & 1 & -3/5 & (b-2a)/5 \\ 0 & 0 & 1 & 9a-2b-5c \end{bmatrix} \quad -5\,R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & -36a+8b+21c \\ 0 & 1 & 0 & 5a-b-3c \\ 0 & 0 & 1 & 9a-2b-5c \end{bmatrix} \quad \begin{matrix} R_1 - \frac{21}{5}R_3 \\ R_2 + \frac{3}{5}R_3 \end{matrix}$$

So, $k_1 = -36a + 8b + 21c, \quad k_2 = 5a - b - 3c, \quad k_3 = 9a - 2b - 5c$

Hence, v_1, v_2, v_3 forms basis for $\mathbb{R}^3$

Example: Let $\boldsymbol{v_1 = (1,1), v_2 = (3,5), v_3 = (4,2)}$. whether v_1, v_2, v_3 form basis for R^2 or not?

Linearly independent or not:

$$k_1v_1 + k_2v_2 + k_3v_3 = 0$$

$$k_1(1,1) + k_2(3,5) + k_3(4,2) = (0,0)$$

$$(k_1 + 3k_2 + 4k_3, k_1 + 5k_2 + 2k_3) = (0,0)$$

$$\left(\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 1 & 5 & 2 & 0 \end{array}\right)$$

By using Gaussian Jordan method, we come up with

$$k_1 = -7k_3, \qquad k_2 = k_3$$

As k_1, k_2, k_3 are not zero. So, v_1, v_2, v_3 are linearly dependent.

Hence, v_1, v_2, v_3 does not form basis for R^2.

Please see the following MS Word file uploaded on portal for further practice

problems

16. Basis and Dimension

THANKS